CYBERSECURITY INTERNSHIP

WEEK 5 — VAPT TOOLS: PRACTICAL LEARNING

Vulnerability Assessment & Penetration Testing

Intern	Pooja Shivakalimath
Internship	DG Interns Hub — Cybersecurity Internship
Authorized target	http://testphp.vulnweb.com
Tools	Nmap • Nikto • Burp Suite

Practical evidence • Reconnaissance • HTTP inspection • Security observations

1. Overview

This week focused on Vulnerability Assessment and Penetration Testing (VAPT) using the authorized training target testphp.vulnweb.com. The practical work covered Nmap reconnaissance, service/version detection, aggressive scanning, Nikto troubleshooting, and Burp Suite HTTP request inspection.

2. Objectives

- Perform basic reconnaissance using Nmap.
- Identify exposed TCP services and understand scan output.
- Understand the limitations of aggressive OS/service detection.
- Use Burp Suite to capture and inspect HTTP requests.
- Document only evidence-supported observations.

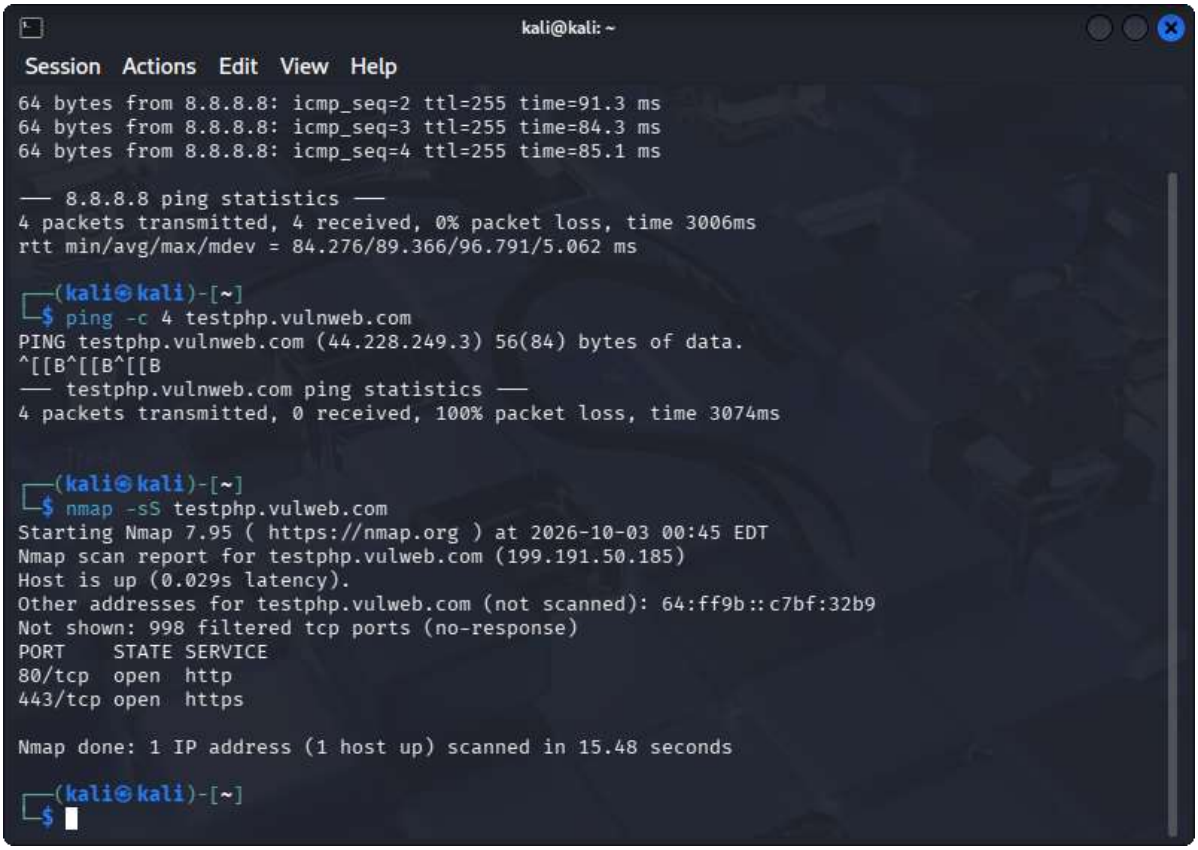
3. Scope & Safety

Testing was limited to the authorized training target supplied for the internship. No unauthorized systems were targeted. Findings below are observations from the captured evidence, not unverified vulnerability claims.

4. Nmap — SYN Scan (-sS)

The SYN scan identified two open web-related TCP ports in the captured result: 80/tcp (HTTP) and 443/tcp (HTTPS).

- 80/tcp — HTTP — open
- 443/tcp — HTTPS — open
- Most other scanned TCP ports were filtered/no-response in the captured result.



5. Nmap — Service/Version Detection (-sV)

The -sV option attempts to identify services and versions. The captured scan completed, but detailed version information was limited by the responses available from the target.



6. Nmap — Aggressive Scan (-A)

The -A scan enabled additional detection features. Nmap explicitly warned that its OS guesses may be unreliable because the required open/closed port conditions were not available. The guesses were therefore not treated as confirmed findings.

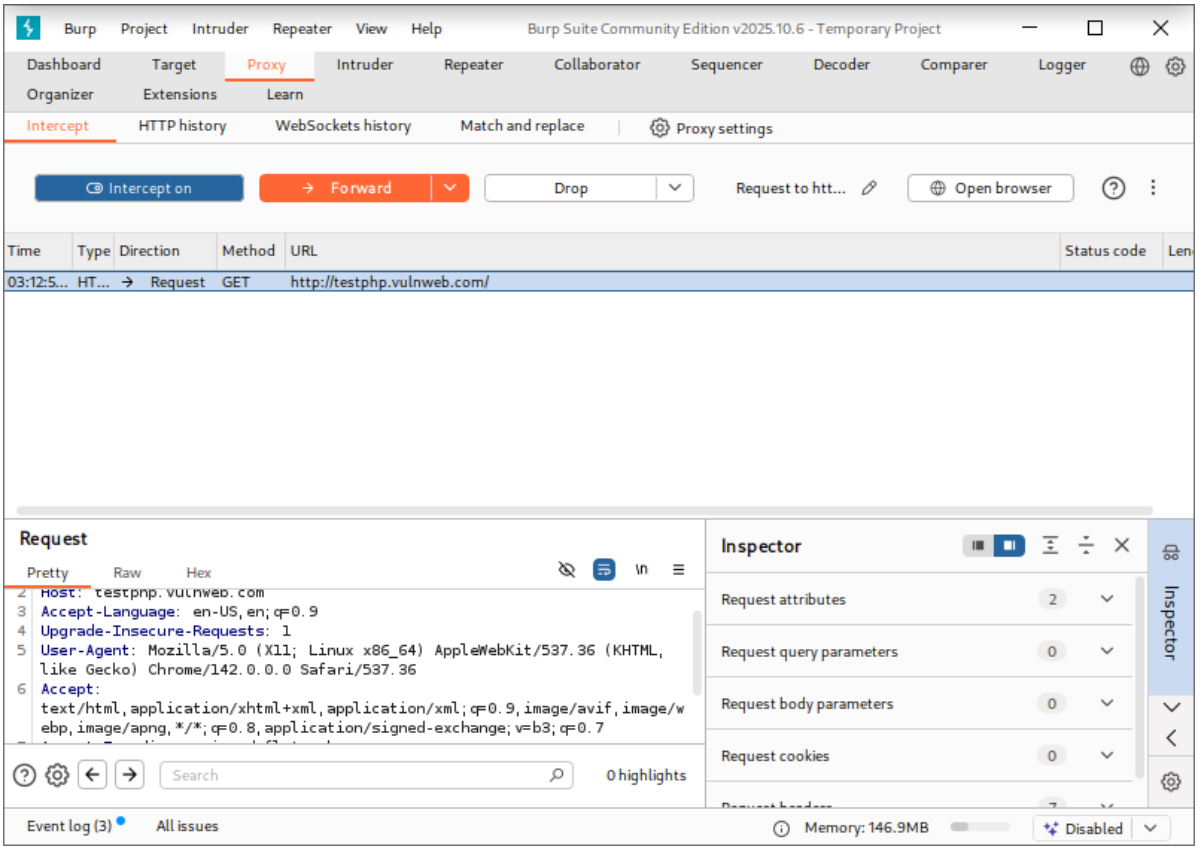


Figure 3 — Nmap aggressive scan (-A)

7. Nikto — Web Server Scanner

Nikto was attempted against the authorized target but could not complete because of connection/name-resolution/proxy-related issues in the Kali environment. No Nikto vulnerability result was invented.

- Tried hostname-based scanning.
- Tried the resolved IP and port 80.
- Tested HTTP connectivity with curl.
- Checked name resolution and proxy configuration.
- Documented the limitation after repeated connection failures.

8. Burp Suite — HTTP Request Inspection

Burp Suite was used as an intercepting proxy on 127.0.0.1:8080. The built-in browser accessed the training site and Burp successfully captured GET requests and HTTP headers including Host, User-Agent, Accept-Language and Accept.

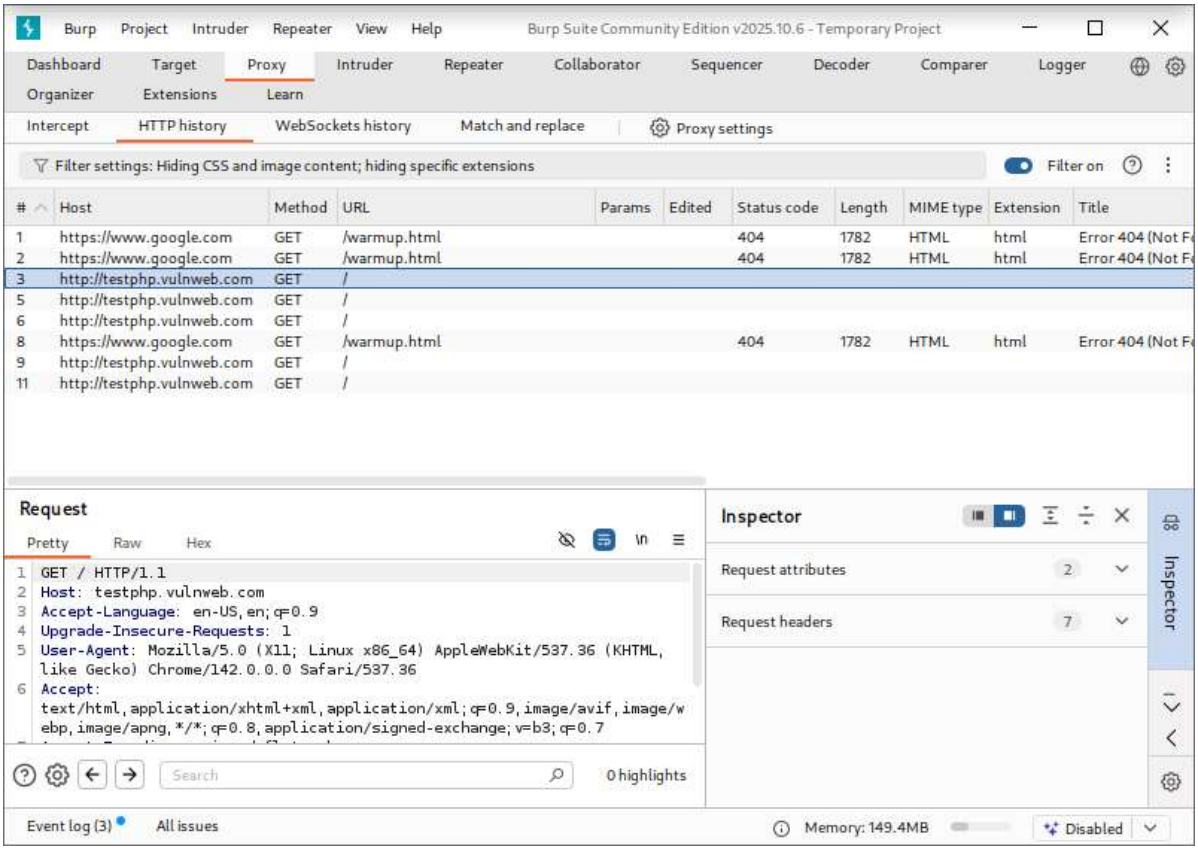


Figure 4 — Burp Suite intercepted HTTP request

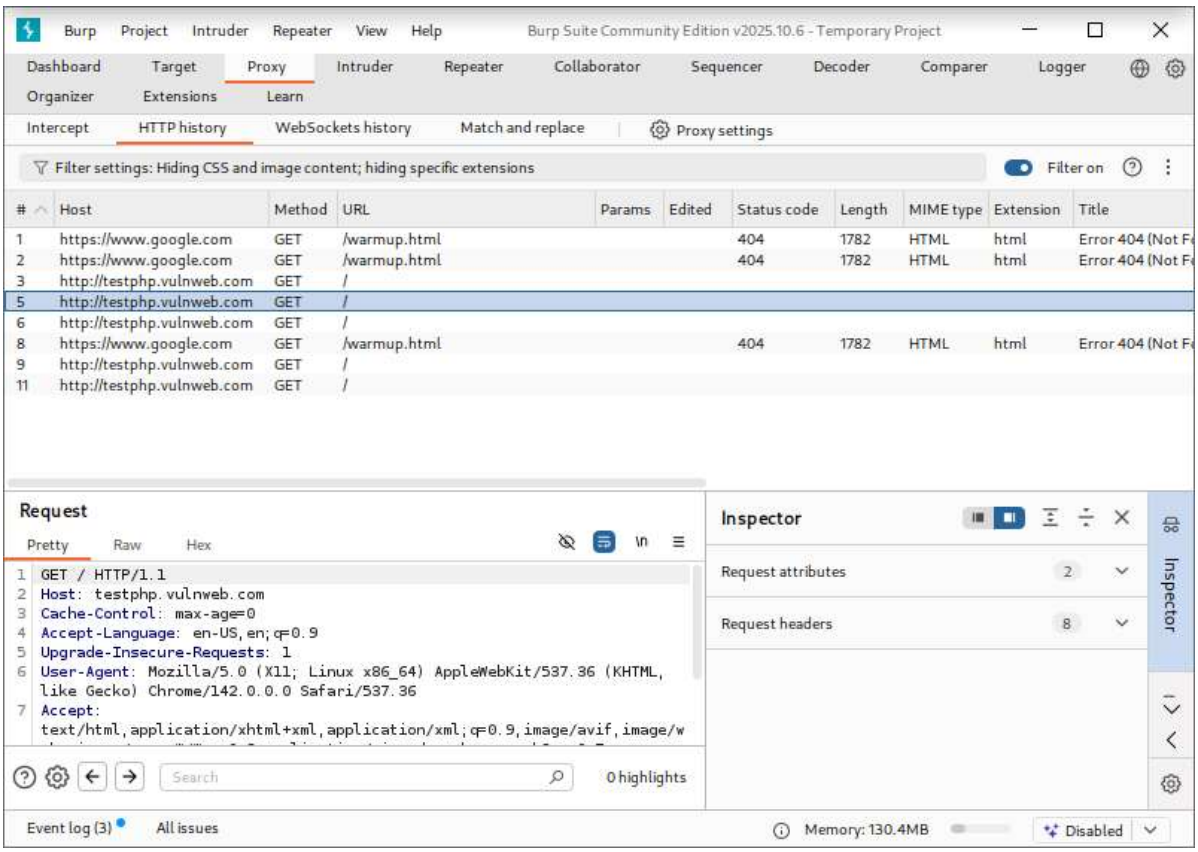


Figure 5 — Burp Suite HTTP history

9. Three Evidence-Based Security Observations

#	Observation	Evidence	Meaning
1	HTTP exposed on 80/tcp	Nmap -sS	Web traffic is available over HTTP; HTTPS should be preferred where appropriate.
2	HTTPS exposed on 443/tcp	Nmap -sS	Encrypted web communication is available; TLS configuration should still be reviewed.
3	HTTP requests/headers can be inspected	Burp Suite	Authorized proxy testing can reveal request methods, host data and headers.

10. Challenges & Solutions

Challenges included Kali networking, Nikto connectivity, and VirtualBox shared-folder permissions. These were handled through step-by-step troubleshooting, AI guidance, controlled tests, and configuration checks. The Nikto issue was documented honestly rather than creating unsupported findings.

11. Learning & Conclusion

- Nmap can be used in different modes for different assessment goals.
- Filtered ports can reduce confidence in service and OS detection.
- Burp Suite provides visibility into browser-to-server HTTP communication.
- Security reports should distinguish confirmed evidence from assumptions.
- Troubleshooting and accurate documentation are important parts of practical cybersecurity.

Overall, Week 5 provided hands-on exposure to the VAPT workflow: reconnaissance, service identification, web assessment, and HTTP traffic inspection.